

Problem Solving Using Python (CSE1012)

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Branching Statement

- Determines the control flow of a program
- Three fundamental methods of control flow:
 - Sequential
 - Selection
 - Iterative control

Branching Statement

- A ***control statement*** is a statement that determines the control flow of a set of instructions
- i.e., it decides the sequence in which the instructions in a program are to be executed.

Branching Statement

- **Sequential Control:** A Python program is executed sequentially from the first line of the program to its last line.
- **Selection Control:** To execute only a selected set of statements.
- **Iterative Control:** To execute a set of statements repeatedly.

Branching Statement

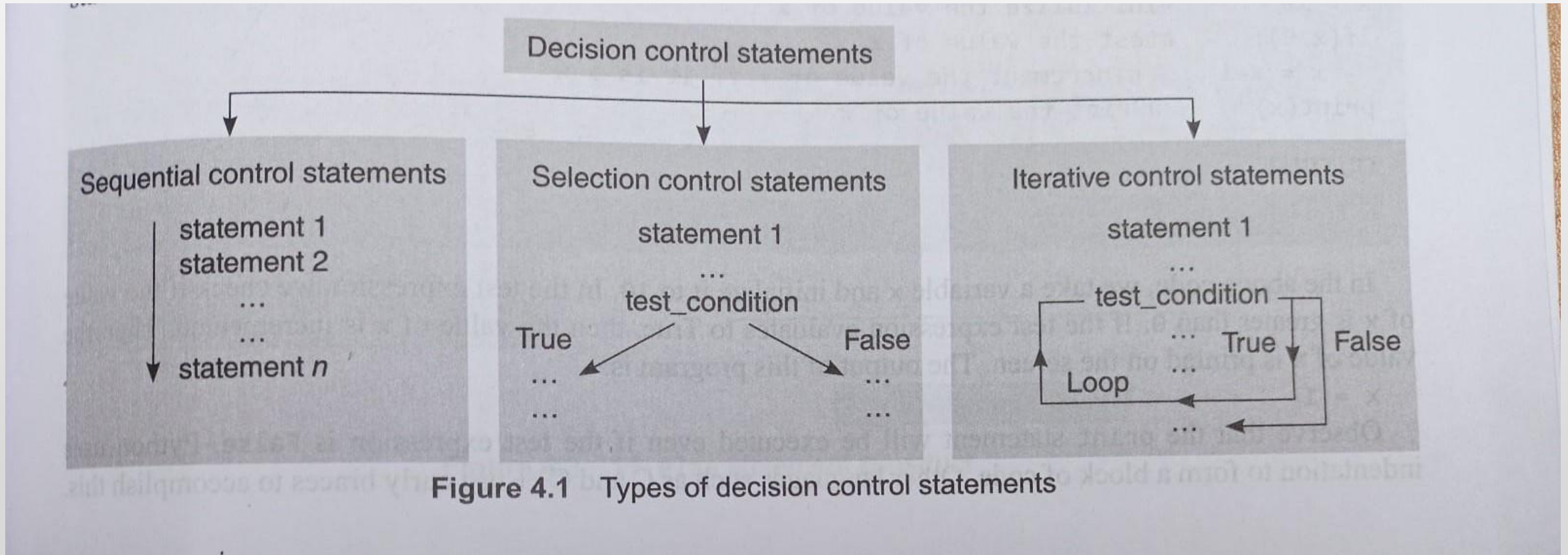


Figure 4.1 Types of decision control statements

Selection/Cond Branching Statement

- Jumps from one part of the program to another
- Depending on whether a particular condition is satisfied or not
- Python supports different types of conditional branching statements:
 - If statement
 - If-else statement
 - Nested If statement
 - If-elif-else statement

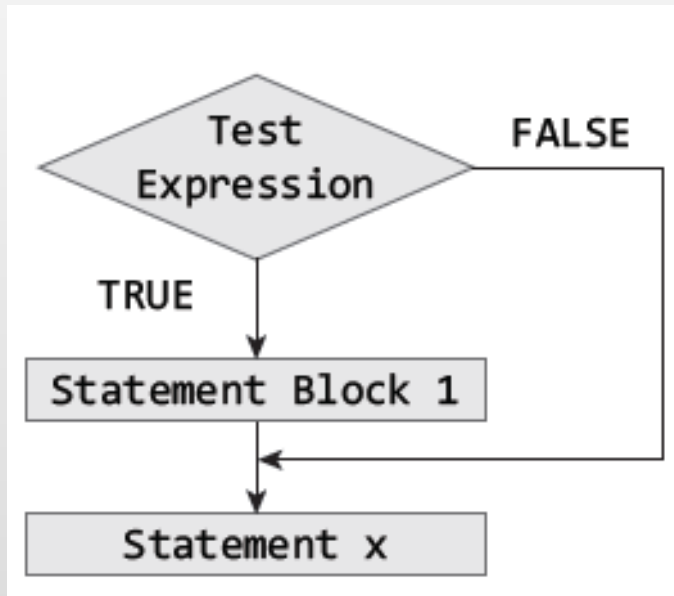
If Statement

- Simplest form of decision control
- Frequently used
- It may include 1 or n number of statements enclosed within If block
- First the 'test expression' is evaluated
- If the 'test expression' is true, the statements of If block are executed
- Otherwise skipped

If Statement

SYNTAX OF IF STATEMENT

```
if test_expression:  
    statement1  
    .....  
    Statement n  
statement x;
```



If Statement

```
x = 10          #Initialize the value of x
if(x>0):        #test the value of x
    x = x+1      #Increment the value of x if it is > 0
print(x)        #Print the value of x
```

OUTPUT

```
x = 11
```

If Statement Example

- # Enter a number, if it is greater than 0 increment by 1

```
x= int(input("Enter a Number: "))
```

```
if(x>0):
```

```
    x=x+1
```

```
print(x)
```

If Statement

- In the above example, the `print()` statement will be executed even if the condition is `False`
- Python uses indentation to form block of code

If Statement Example

- # Determine if a person is eligible to Vote

```
age = int(input("Enter the age: "))
```

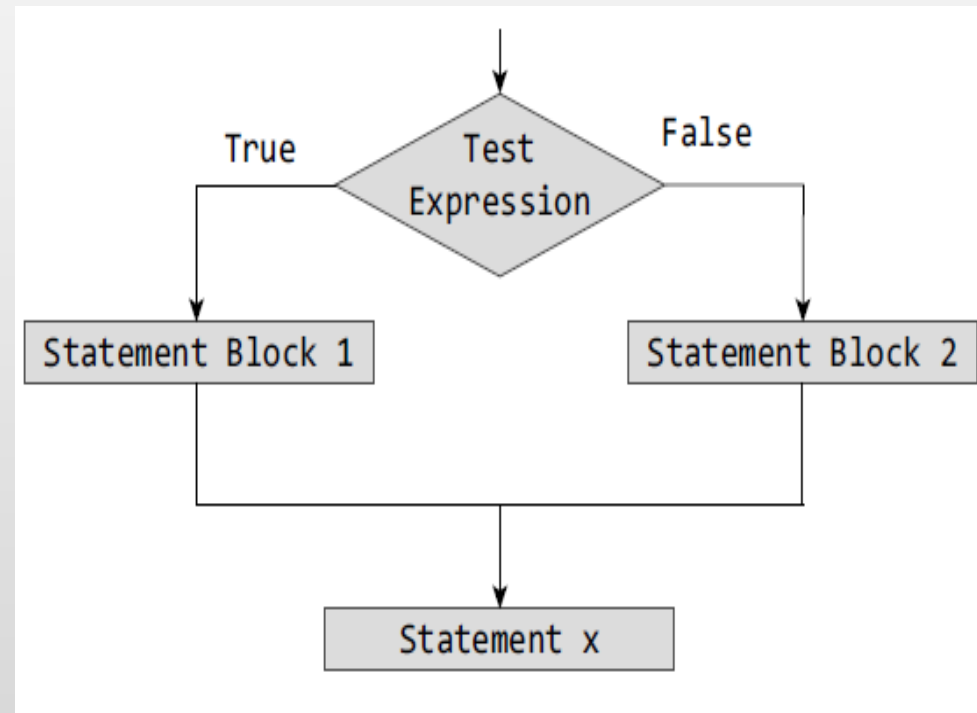
```
if(age >= 18):
```

```
    print("You are eligible to vote")
```

If-else Statement

SYNTAX OF IF-ELSE STATEMENT

```
if (test expression):  
    statement block 1  
else:  
    statement block 2  
statement x;
```



If-else Statement

```
age = int(input("Enter the age : "))  
if(age>=18):  
    print("You are eligible to vote")  
else:  
    yrs = 18 - age  
    print("You have to wait for another " + str(yrs) +" years to cast your vote")
```

OUTPUT

```
Enter the age : 10  
You have to wait for another 8 years to cast your vote
```

If-else Statement Example

- # Determine the larger of the two numbers

If-else Statement Example

- # Determine the larger of the two numbers

```
a = int(input("Enter the value of a: "))
```

```
b = int(input("Enter the value of b: "))
```

```
if(a > b):
```

```
    large = a
```

```
else:
```

```
    large = b
```

```
print("Large= ",large)
```

If-else Statement Example

- # Determine if a number is even or Odd

If-else Statement Example

- # Determine if a number is even or Odd

```
num = int(input("Enter the number: "))
```

```
if(num % 2 == 0):
```

```
    print(num, "is even")
```

```
else:
```

```
    print(num, "is odd")
```

Nested If Statements

- An If statement inside another If statement
- Can be used to check multiple conditions subject to the satisfaction of the first condition

Nested If Statements

#Example of Nested If statement

```
num=int(input("Enter a number: "))
```

```
if(num>=0):
```

```
    if(num<=30):
```

```
        print(num,"is in the range 0-30")
```

Nested If Statements

```
num = int(input("Enter any number from 0-30: "))
if(num>=0 and num<10):
    print("It is in the range 0-10")
elif(num>=10 and num<20):
    print("It is in the range 10-20")
elif(num>=20 and num<30):
    print("It is in the range 20-30")
```

OUTPUT

```
Enter any number from 0-30: 25
It is in the range 20-30
```

If-elif-else Statements

- Python supports if-elif-else statements to test additional conditions apart from the initial test expression.
- The if-elif-else construct works in the same way as a usual if-else statement.
- If-elif-else construct is also known as **nested-if** construct.

If-elif-else Statements

```
num = int(input("Enter any number : "))  
if(num==0):  
    print("The value is equal to zero")  
elif(num>0):  
    print("The number is positive")  
else:  
    print("The number is negative")
```

OUTPUT

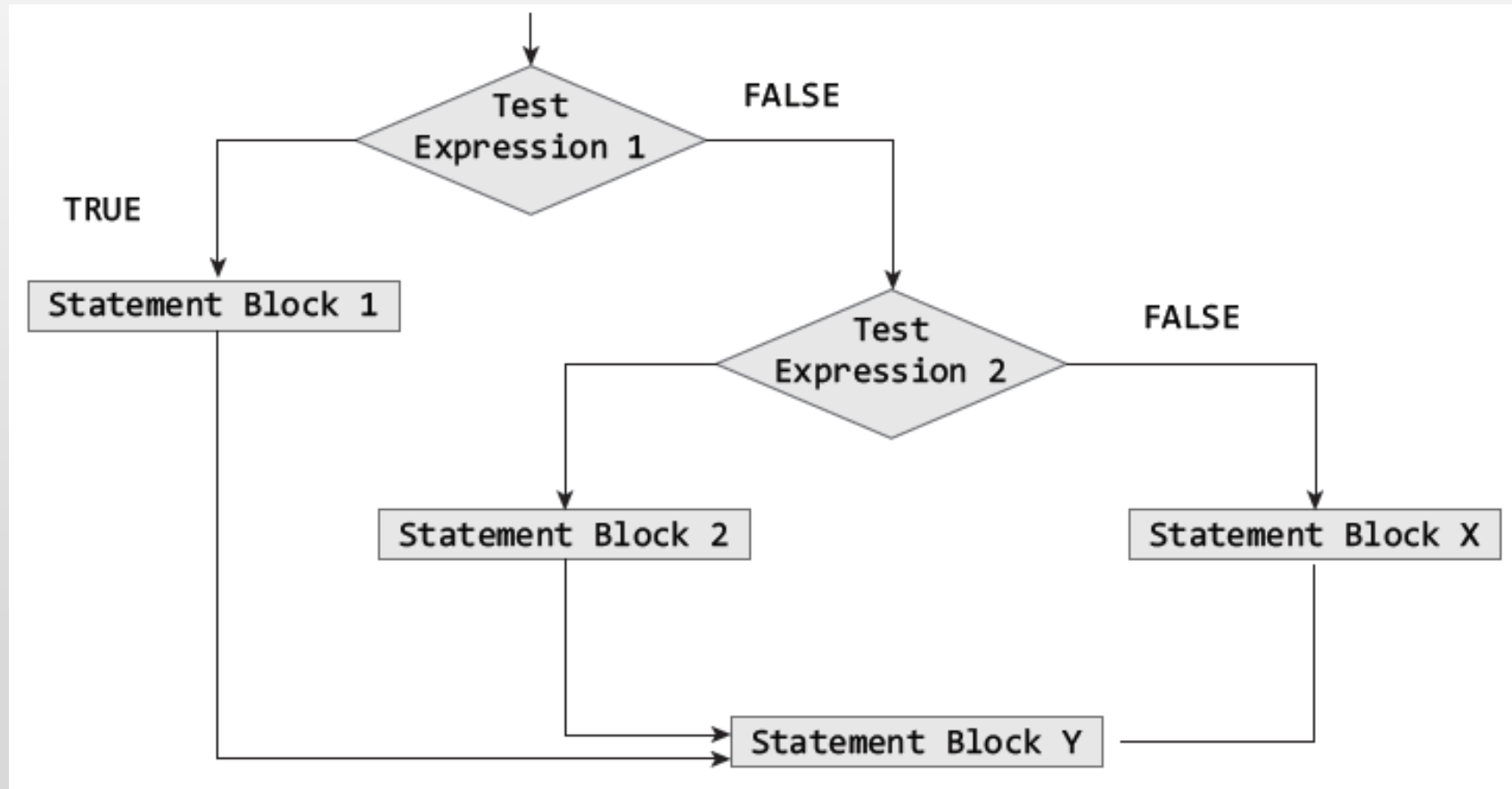
```
Enter any number : -10  
The number is negative
```


If-elif-else Statements

Calculation of Student Grade

```
number = int(input("Enter the numeric grade: "))  
if number > 89:  
    letter = 'A'  
elif number > 79:  
    letter = 'B'  
elif number > 69:  
    letter = 'C'  
else:  
    letter = 'F'  
print("The letter grade is", letter)
```

If-elif-else Statements



Home Work

1. Write an algorithm to determine greatest of three numbers.
Write corresponding Python code
2. Write an algorithm to determine a char entered is a vowel or consonant. Write Python code for it.

Home Work

3. Write an algorithm to take an input from user and then check whether it is a number or a character. Write corresponding Python code.
4. Write an algorithm to calculate roots of a quadratic equation. Write Python code for it.

Thank You

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